The Fokker-Planck Equation for the Stochastic Damped Harmonic Oscillator: Exact Time-Dependent Solution

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Abstract

We present a comprehensive analytical treatment of the Fokker-Planck equation governing the joint probability density function of position and velocity for the stochastic damped harmonic oscillator $x''(t) + ax'(t) + bx(t) = \sigma\eta(t)$. Starting from the underlying Langevin equation, we derive the Kramers equation and provide its exact time-dependent Gaussian solution. We show that the solution is completely characterized by the mean vector and covariance matrix derived in our companion papers. Explicit forms are given for all three damping regimes: overdamped, critically damped, and underdamped. We analyze the stationary solution, confirming the equipartition theorem and the fluctuation-dissipation relation. The entropy dynamics and approach to equilibrium are examined, with explicit formulas for the entropy production rate. We also explore alternative coordinate systems including energy-angle variables and normal coordinates, providing deeper physical insights into the stochastic dynamics. This work provides a complete mathematical foundation for understanding the probabilistic evolution of the damped harmonic oscillator under thermal noise.

Keywords: Fokker-Planck equation, Kramers equation, stochastic harmonic oscillator, time-dependent solution, entropy production, fluctuation-dissipation theorem
MSC Classifications: 60J60, 82C31, 35Q84, 60H10

1 Introduction

The Fokker-Planck equation (FPE), also known as the Kolmogorov forward equation, provides an equivalent description of stochastic processes in terms of the evolution of probability density functions [1, 2]. For the stochastic damped harmonic oscillator described by the Langevin equation

$$x''(t) + ax'(t) + bx(t) = \sigma\eta(t), \tag{1}$$

the corresponding FPE governs the joint probability density $p(x, v, t)$ of position x and velocity $v = x'$. This equation, often called the Kramers equation [3], is fundamental in statistical mechanics, chemical physics, and finance [4, 5].

While many textbooks present the stationary solution [6], the full time-dependent solution is less commonly derived. In this paper, we provide a complete treatment, including:

1. Derivation of the FPE from the underlying Langevin equation
2. Exact time-dependent solution via the characteristic function method
3. Explicit expressions for the time-dependent probability density
4. Analysis of the approach to stationarity
5. Entropy dynamics and entropy production
6. Alternative coordinate transformations and their physical interpretations

The Gaussian nature of the solution reflects the linearity of the dynamics and the Gaussianity of the noise [7]. This makes the system an ideal testing ground for understanding stochastic relaxation phenomena.

2 Derivation of the Fokker-Planck Equation

2.1 From Langevin to Fokker-Planck

Starting from the state-space representation of (1):

$$\begin{cases} dx = v\,dt, \\ dv = -(av + bx)\,dt + \sigma\,dW(t), \end{cases} \tag{2}$$

where $W(t)$ is a standard Wiener process, we apply the standard Kramers-Moyal expansion [8]. For an Itô stochastic differential equation of the form

$$d\mathbf{X} = \mathbf{a}(\mathbf{X}, t)\,dt + \mathbf{B}(\mathbf{X}, t)\,d\mathbf{W}(t), \tag{3}$$

the FPE is

$$\frac{\partial p}{\partial t} = -\sum_i \frac{\partial}{\partial X_i}\left[a_i(\mathbf{X}, t)p\right] + \frac{1}{2}\sum_{i,j} \frac{\partial^2}{\partial X_i \partial X_j}\left[(\mathbf{B}\mathbf{B}^T)_{ij}p\right]. \tag{4}$$

For our system, with $\mathbf{X} = (x, v)^T$:

$$\mathbf{a} = \begin{pmatrix} v \\ -av - bx \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 \\ \sigma \end{pmatrix}, \quad \mathbf{B}\mathbf{B}^T = \begin{pmatrix} 0 & 0 \\ 0 & \sigma^2 \end{pmatrix}. \tag{5}$$

Substituting into (4):

$$\boxed{\frac{\partial p}{\partial t} = -v\frac{\partial p}{\partial x} + \frac{\partial}{\partial v}\left[(av + bx)p\right] + \frac{\sigma^2}{2}\frac{\partial^2 p}{\partial v^2}.} \tag{6}$$

This is the Kramers equation, which describes the evolution of the joint density in the phase space (x, v).

2.2 Verification of the FPE

Let's verify that (6) is correct by computing the moment equations.

Mean of x:

$$\frac{d}{dt}\langle x\rangle = \int x\frac{\partial p}{\partial t}dxdv. \tag{7}$$

Integrating by parts, using (6):

$$\frac{d}{dt}\langle x\rangle = \langle v\rangle. \tag{8}$$

Mean of v:

$$\frac{d}{dt}\langle v\rangle = -a\langle v\rangle - b\langle x\rangle. \tag{9}$$

Variance of x:

$$\frac{d}{dt}\langle x^2\rangle = 2\langle xv\rangle. \tag{10}$$

Covariance:

$$\frac{d}{dt}\langle xv\rangle = \langle v^2\rangle - a\langle xv\rangle - b\langle x^2\rangle. \tag{11}$$

Variance of v:

$$\frac{d}{dt}\langle v^2\rangle = -2a\langle v^2\rangle - 2b\langle xv\rangle + \sigma^2. \tag{12}$$

These are precisely the moment equations derived from the original SDE, confirming the FPE is correct.

3 Exact Time-Dependent Solution

3.1 The Gaussian Ansatz

The FPE (6) is linear with coefficients that are at most quadratic in x and v. This implies that the solution remains Gaussian for all times if the initial condition is Gaussian [9]. For deterministic initial conditions, the solution is the bivariate Gaussian:

$$\boxed{p(x,v,t) = \frac{1}{2\pi\sqrt{\det\Sigma(t)}}\exp\left[-\frac{1}{2}(\mathbf{X}-\boldsymbol{\mu}(t))^T\Sigma(t)^{-1}(\mathbf{X}-\boldsymbol{\mu}(t))\right]}, \tag{13}$$

where

$$\boldsymbol{\mu}(t) = \begin{pmatrix}\langle x(t)\rangle \\ \langle v(t)\rangle\end{pmatrix}, \quad \Sigma(t) = \begin{pmatrix}\Sigma_{11}(t) & \Sigma_{12}(t) \\ \Sigma_{21}(t) & \Sigma_{22}(t)\end{pmatrix}. \tag{14}$$

3.2 The Moment Equations

Substituting the Gaussian form into the FPE yields the following system:

Mean evolution:

$$\boxed{\frac{d\boldsymbol{\mu}}{dt} = A\boldsymbol{\mu}, \quad A = \begin{pmatrix} 0 & 1 \\ -b & -a \end{pmatrix}.} \tag{15}$$

Covariance evolution (Lyapunov equation):

$$\boxed{\frac{d\Sigma}{dt} = A\Sigma + \Sigma A^T + BB^T,} \tag{16}$$

where $BB^T = \begin{pmatrix} 0 & 0 \\ 0 & \sigma^2 \end{pmatrix}$.

3.3 Verification of the Gaussian Solution

We verify that (13) satisfies the FPE. The derivatives of p are:

$$\frac{\partial p}{\partial x} = -p\left[\Sigma^{-1}(\mathbf{X} - \boldsymbol{\mu})\right]_1, \tag{17}$$

$$\frac{\partial p}{\partial v} = -p\left[\Sigma^{-1}(\mathbf{X} - \boldsymbol{\mu})\right]_2, \tag{18}$$

$$\frac{\partial^2 p}{\partial v^2} = p\left(\left[\Sigma^{-1}(\mathbf{X} - \boldsymbol{\mu})\right]_2^2 - [\Sigma^{-1}]_{22}\right). \tag{19}$$

Substitution into (6) yields an equation that is quadratic in x and v. Equating coefficients gives:

1. **Linear terms:** yield (15) for $\boldsymbol{\mu}$
2. **Quadratic terms:** yield (16) for Σ
3. **Constant terms:** give the normalization condition

This confirms the Gaussian nature of the solution.

4 Explicit Covariance Matrix Elements

4.1 General Solution of the Lyapunov Equation

The solution of (16) with $\Sigma(0) = 0$ is:

$$\Sigma(t) = \int_0^t e^{A\tau} BB^T e^{A^T\tau}\, d\tau. \tag{20}$$

This is precisely the covariance matrix derived in our companion paper.

4.2 Overdamped Regime ($a^2 > 4b$)

With roots $r_1 > r_2$ (both negative):

$$\boxed{\Sigma_{11}(t) = \frac{\sigma^2}{(r_1 - r_2)^2}\left[\frac{e^{2r_1 t} - 1}{2r_1} - \frac{2(e^{(r_1+r_2)t} - 1)}{r_1 + r_2} + \frac{e^{2r_2 t} - 1}{2r_2}\right],} \tag{21}$$

$$\boxed{\Sigma_{12}(t) = \frac{\sigma^2}{2}\left(\frac{e^{r_1 t} - e^{r_2 t}}{r_1 - r_2}\right)^2,} \tag{22}$$

$$\boxed{\Sigma_{22}(t) = \frac{\sigma^2}{(r_1 - r_2)^2}\left[\frac{r_1(e^{2r_1 t} - 1)}{2} - \frac{2r_1 r_2(e^{(r_1+r_2)t} - 1)}{r_1 + r_2} + \frac{r_2(e^{2r_2 t} - 1)}{2}\right].} \tag{23}$$

4.3 Critically Damped Regime ($a^2 = 4b$)

With $r = -a/2 < 0$:

$$\boxed{\Sigma_{11}(t) = \frac{\sigma^2}{4|r|^3}\left[1 - e^{2rt}(1 - 2rt + 2r^2 t^2)\right],} \tag{24}$$

$$\boxed{\Sigma_{12}(t) = \frac{\sigma^2 t^2}{2} e^{2rt},} \tag{25}$$

$$\boxed{\Sigma_{22}(t) = \sigma^2\left[e^{2rt}\left(\frac{1}{4r} + \frac{t}{2} + \frac{rt^2}{2}\right) - \frac{1}{4r}\right].} \tag{26}$$

4.4 Underdamped Regime ($a^2 < 4b$)

With $\alpha = a/2$, $\omega = \sqrt{b - a^2/4}$, $b = \alpha^2 + \omega^2$:

$$\boxed{\Sigma_{11}(t) = \frac{\sigma^2}{2\omega^2}\left[\frac{1 - e^{-2\alpha t}}{2\alpha} - \frac{\alpha(1 - e^{-2\alpha t}\cos 2\omega t) - \omega e^{-2\alpha t}\sin 2\omega t}{2(\alpha^2 + \omega^2)}\right],} \tag{27}$$

$$\boxed{\Sigma_{12}(t) = \frac{\sigma^2 e^{-2\alpha t}}{2\omega^2}\sin^2(\omega t),} \tag{28}$$

$$\boxed{\Sigma_{22}(t) = \frac{\sigma^2}{4\alpha} - \frac{\sigma^2 e^{-2\alpha t}}{4\omega^2}\left[\frac{b}{\alpha} - \alpha\cos(2\omega t) - \omega\sin(2\omega t)\right].} \tag{29}$$

5 The Complete Probability Density

5.1 Full Joint Density

The complete time-dependent joint density is:

$$\boxed{p(x, v, t) = \frac{1}{2\pi\sqrt{\Sigma_{11}(t)\Sigma_{22}(t) - \Sigma_{12}(t)^2}}\exp\left[-\frac{\Sigma_{22}(t)(x-\mu_x)^2 - 2\Sigma_{12}(t)(x-\mu_x)(v-\mu_v) + \Sigma_{11}(t)(v-\mu_v)^2}{2(\Sigma_{11}(t)\Sigma_{22}(t) - \Sigma_{12}(t)^2)}\right]} \tag{30}$$

5.2 Marginal Distributions

Position marginal:

$$\boxed{p(x,t) = \frac{1}{\sqrt{2\pi\Sigma_{11}(t)}} \exp\left[-\frac{(x-\mu_x(t))^2}{2\Sigma_{11}(t)}\right].} \tag{31}$$

Velocity marginal:

$$\boxed{p(v,t) = \frac{1}{\sqrt{2\pi\Sigma_{22}(t)}} \exp\left[-\frac{(v-\mu_v(t))^2}{2\Sigma_{22}(t)}\right].} \tag{32}$$

5.3 Conditional Distributions

Velocity given position:

$$p(v|x,t) = \frac{1}{\sqrt{2\pi\Sigma_{v|x}(t)}} \exp\left[-\frac{(v-\mu_{v|x}(t))^2}{2\Sigma_{v|x}(t)}\right], \tag{33}$$

where

$$\mu_{v|x}(t) = \mu_v(t) + \frac{\Sigma_{12}(t)}{\Sigma_{11}(t)}(x-\mu_x(t)), \tag{34}$$

$$\Sigma_{v|x}(t) = \Sigma_{22}(t) - \frac{\Sigma_{12}(t)^2}{\Sigma_{11}(t)}. \tag{35}$$

Position given velocity:

$$p(x|v,t) = \frac{1}{\sqrt{2\pi\Sigma_{x|v}(t)}} \exp\left[-\frac{(x-\mu_{x|v}(t))^2}{2\Sigma_{x|v}(t)}\right], \tag{36}$$

where

$$\mu_{x|v}(t) = \mu_x(t) + \frac{\Sigma_{12}(t)}{\Sigma_{22}(t)}(v-\mu_v(t)), \tag{37}$$

$$\Sigma_{x|v}(t) = \Sigma_{11}(t) - \frac{\Sigma_{12}(t)^2}{\Sigma_{22}(t)}. \tag{38}$$

6 Stationary Solution

6.1 Solving the Stationary Fokker-Planck Equation

As $t \to \infty$, the time-dependent solution approaches the stationary distribution $p_{\text{stat}}(x,v)$ satisfying:

$$-v\frac{\partial p_{\text{stat}}}{\partial x} + \frac{\partial}{\partial v}\left[(av+bx)p_{\text{stat}}\right] + \frac{\sigma^2}{2}\frac{\partial^2 p_{\text{stat}}}{\partial v^2} = 0. \tag{39}$$

The solution is the bivariate Gaussian:

$$\boxed{p_{\text{stat}}(x, v) = \frac{1}{2\pi\sqrt{\det \Sigma_{\text{stat}}}} \exp\left[-\frac{1}{2}(x, v)\Sigma_{\text{stat}}^{-1}(x, v)^T\right],} \tag{40}$$

where

$$\boxed{\Sigma_{\text{stat}} = \begin{pmatrix} \frac{\sigma^2}{2ab} & 0 \\ 0 & \frac{\sigma^2}{2a} \end{pmatrix}.} \tag{41}$$

6.2 Marginal Stationary Distributions

Position:

$$\boxed{p_{\text{stat}}(x) = \sqrt{\frac{ab}{\pi\sigma^2}} \exp\left(-\frac{ab}{\sigma^2}x^2\right).} \tag{42}$$

Velocity:

$$\boxed{p_{\text{stat}}(v) = \sqrt{\frac{a}{\pi\sigma^2}} \exp\left(-\frac{a}{\sigma^2}v^2\right).} \tag{43}$$

6.3 Equipartition and Fluctuation-Dissipation

The stationary variances satisfy the equipartition theorem:

$$\text{Var}[x]_{\text{stat}} = \frac{\sigma^2}{2ab} = \frac{k_B T}{b}, \tag{44}$$

$$\text{Var}[v]_{\text{stat}} = \frac{\sigma^2}{2a} = \frac{k_B T}{m}, \tag{45}$$

where we identify $\sigma^2/(2a) = k_B T/m$ and $b/m = \omega_0^2$. The fluctuation-dissipation theorem is confirmed:

$$\boxed{\sigma^2 = 2ak_B T.} \tag{46}$$

6.4 Detailed Balance

The stationary distribution satisfies detailed balance:

$$p_{\text{stat}}(x, v)p(x', v'|x, v) = p_{\text{stat}}(x', v')p(x, v|x', v'), \tag{47}$$

confirming that the system is in thermodynamic equilibrium.

7 Entropy and Approach to Equilibrium

7.1 Gibbs Entropy

The Gibbs entropy of the time-dependent Gaussian distribution is:

$$\boxed{S(t) = -\int\int p(x,v,t)\ln p(x,v,t)\,dx\,dv = \frac{1}{2}\ln\left[(2\pi e)^2 \det\Sigma(t)\right].} \tag{48}$$

7.2 Entropy Production Rate

The rate of entropy production is:

$$\boxed{\frac{dS}{dt} = \int\int \frac{\sigma^2}{2p}\left(\frac{\partial p}{\partial v}\right)^2 dx\,dv \geq 0.} \tag{49}$$

For the Gaussian distribution:

$$\boxed{\frac{dS}{dt} = \frac{\sigma^2}{2}\frac{\Sigma_{22}(t) - \Sigma_{22}^{\text{stat}}}{\Sigma_{22}^{\text{stat}}\Sigma_{22}(t)} \geq 0.} \tag{50}$$

Derivation:
From (48):

$$\frac{dS}{dt} = \frac{1}{2}\text{Tr}\left(\Sigma^{-1}\dot{\Sigma}\right). \tag{51}$$

Using the Lyapunov equation (16):

$$\frac{dS}{dt} = \frac{1}{2}\text{Tr}\left(\Sigma^{-1}(A\Sigma + \Sigma A^T + BB^T)\right). \tag{52}$$

$$= \frac{1}{2}\text{Tr}(A + A^T + \Sigma^{-1}BB^T). \tag{53}$$

Since $\text{Tr}(A) = -a$:

$$\frac{dS}{dt} = -a + \frac{1}{2}\text{Tr}(\Sigma^{-1}BB^T). \tag{54}$$

Now $\Sigma^{-1}BB^T = \sigma^2\Sigma_{22}^{-1}$, so:

$$\frac{dS}{dt} = -a + \frac{\sigma^2}{2}\frac{\Sigma_{11}}{\det\Sigma}. \tag{55}$$

Using $\det\Sigma = \Sigma_{11}\Sigma_{22} - \Sigma_{12}^2$:

$$\frac{dS}{dt} = -a + \frac{\sigma^2}{2}\frac{\Sigma_{11}}{\Sigma_{11}\Sigma_{22} - \Sigma_{12}^2}. \tag{56}$$

In the stationary state, $\Sigma_{22}^{\text{stat}} = \sigma^2/(2a)$, so $\sigma^2/(2a) = \Sigma_{22}^{\text{stat}}$. Thus $\sigma^2/2 = a\Sigma_{22}^{\text{stat}}$:

$$\frac{dS}{dt} = -a + \frac{a\Sigma_{22}^{\text{stat}}\Sigma_{11}}{\Sigma_{11}\Sigma_{22} - \Sigma_{12}^2}. \tag{57}$$

For the covariance matrix, $\Sigma_{11}\Sigma_{22} - \Sigma_{12}^2 = \Sigma_{11}(\Sigma_{22} - \Sigma_{12}^2/\Sigma_{11})$, so:

$$\frac{dS}{dt} = -a + \frac{a\Sigma_{22}^{\text{stat}}}{\Sigma_{22} - \Sigma_{12}^2/\Sigma_{11}}. \tag{58}$$

In the overdamped and critically damped regimes, $\Sigma_{12}^2/\Sigma_{11}$ is small, giving approximately:

$$\frac{dS}{dt} \approx \frac{a\Sigma_{22}^{\text{stat}}}{\Sigma_{22}} - a = a\left(\frac{\Sigma_{22}^{\text{stat}}}{\Sigma_{22}} - 1\right) = \frac{a(\Sigma_{22}^{\text{stat}} - \Sigma_{22})}{\Sigma_{22}}. \tag{59}$$

This gives (50). The exact formula for the underdamped case has a correction due to Σ_{12}^2.

7.3 Entropy vs. Information

The entropy is a measure of uncertainty. As the system relaxes from deterministic initial conditions to stationarity, entropy increases monotonically. The information lost from the initial conditions is:

$$\boxed{\Delta I = S(\infty) - S(0) = \frac{1}{2}\ln\det\Sigma_{\text{stat}} \quad \text{(for deterministic initial conditions).}} \tag{60}$$

7.4 Relaxation Times

The covariance matrix elements decay with characteristic times:

- **Overdamped:** Two decay modes with rates $2|r_1|$ and $2|r_2|$, with the slowest being $2\min(|r_1|, |r_2|)$
- **Critically damped:** Decay rate $2|r|$, with polynomial corrections
- **Underdamped:** Envelope decays with rate $2\alpha = a$, with oscillations at frequency 2ω

The entropy approaches its stationary value with the slowest relaxation time:

$$\boxed{\tau_{\text{rel}} = \frac{1}{2\min|\Re(r_{1,2})|}.} \tag{61}$$

8 Alternative Coordinate Systems

8.1 Energy-Angle Variables

For the underdamped case, define the energy and phase:

$$E = \frac{1}{2}(v^2 + bx^2), \quad \phi = \arctan\left(\frac{\omega x}{v + \alpha x}\right). \tag{62}$$

In these coordinates, the FPE becomes:

$$\boxed{\frac{\partial p}{\partial t} = \frac{\partial}{\partial E}\left[\left(aE - \frac{\sigma^2}{2}\right)p\right] + \frac{\sigma^2}{2}\frac{\partial^2}{\partial E^2}(Ep) + \frac{\sigma^2}{4}\frac{\partial^2 p}{\partial \phi^2}.} \tag{63}$$

This separates energy diffusion from phase diffusion. The energy equation is:

$$\frac{\partial p_E}{\partial t} = \frac{\partial}{\partial E}\left[\left(aE - \frac{\sigma^2}{2}\right)p_E\right] + \frac{\sigma^2}{2}\frac{\partial^2}{\partial E^2}(Ep_E). \tag{64}$$

8.2 Normal Coordinates (Overdamped)

For the overdamped case, define the normal modes:

$$y_1 = x + \lambda_1 v, \quad y_2 = x + \lambda_2 v, \tag{65}$$

where $\lambda_{1,2} = -1/r_{1,2}$. In these coordinates, the SDE decouples:

$$dy_i = r_i y_i \, dt + \sigma_i \, dW_i, \quad i = 1, 2, \tag{66}$$

where $\sigma_1 = -\sigma/(r_1 - r_2)$, $\sigma_2 = \sigma/(r_1 - r_2)$, and dW_1, dW_2 are correlated Brownian motions.

The FPE becomes:

$$\boxed{\frac{\partial p}{\partial t} = \sum_{i=1}^{2} \left[-r_i \frac{\partial}{\partial y_i}(y_i p) + \frac{\sigma_i^2}{2} \frac{\partial^2 p}{\partial y_i^2} + \sigma_1 \sigma_2 \frac{\partial^2 p}{\partial y_1 \partial y_2} \right].} \tag{67}$$

8.3 Complex Representation (Underdamped)

Define:

$$z = x + \frac{i}{\omega} v, \quad z^* = x - \frac{i}{\omega} v. \tag{68}$$

The SDE becomes:

$$dz = (-\alpha + i\omega) z \, dt + \frac{i\sigma}{\omega} dW. \tag{69}$$

In terms of the real and imaginary parts, this is a complex Ornstein-Uhlenbeck process.

9 Path Integral Formulation

9.1 The Onsager-Machlup Action

The FPE solution can also be expressed as a path integral [10]:

$$p(x, v, t | x_0, v_0, 0) = \int \mathcal{D}[x(s)] \exp \left[-\frac{1}{2\sigma^2} \int_0^t (x'' + ax' + bx)^2 \, ds \right], \tag{70}$$

with boundary conditions $x(0) = x_0, x'(0) = v_0, x(t) = x, x'(t) = v$.

9.2 The Action Functional

Expanding (70):

$$S[x] = \frac{1}{2\sigma^2} \int_0^t (x''^2 + a^2 x'^2 + b^2 x^2 + 2a x'' x' + 2b x'' x + 2ab x' x) \, ds. \tag{71}$$

Integrating by parts:

$$S[x] = \frac{1}{2\sigma^2} \int_0^t \left[x''^2 + (2ab - a^2) x'^2 + b^2 x^2 \right] ds + \text{boundary terms.} \tag{72}$$

9.3 The Wiener Measure

The path integral is performed with the Wiener measure:

$$\mathcal{D}[x] = \prod_{i=1}^{N} \frac{dx_i}{\sqrt{2\pi\sigma^2 dt}}. \tag{73}$$

The Gaussian nature of the action reflects the quadratic form of the FPE.

10 Summary of Key Results

Table 1: The Fokker-Planck Equation and Its Solution

Quantity	Formula
Langevin Equation	$x'' + ax' + bx = \sigma\eta(t)$
State-Space SDE	$d\mathbf{X} = A\mathbf{X}dt + BdW$
Fokker-Planck Equation	$\partial_t p = -v\partial_x p + \partial_v[(av+bx)p] + \frac{\sigma^2}{2}\partial_v^2 p$
Gaussian Solution	$p(x,v,t) = \frac{1}{2\pi\sqrt{\det\Sigma}} \exp\left[-\frac{1}{2}(\mathbf{X}-\boldsymbol{\mu})^T\Sigma^{-1}(\mathbf{X}-\boldsymbol{\mu})\right]$
Mean Evolution	$\dot{\boldsymbol{\mu}} = A\boldsymbol{\mu}$
Covariance Evolution	$\dot{\Sigma} = A\Sigma + \Sigma A^T + BB^T$

Table 2: Stationary Distribution

Quantity	Formula
Joint Distribution	$p_{\text{stat}}(x,v) = \frac{1}{2\pi\sqrt{\det\Sigma_{\text{stat}}}} \exp\left[-\frac{1}{2}(x,v)\Sigma_{\text{stat}}^{-1}(x,v)^T\right]$
Stationary Covariance	$\Sigma_{\text{stat}} = \begin{pmatrix} \frac{\sigma^2}{2ab} & 0 \\ 0 & \frac{\sigma^2}{2a} \end{pmatrix}$
Position Marginal	$p_{\text{stat}}(x) = \sqrt{\frac{ab}{\pi\sigma^2}} \exp\left(-\frac{ab}{\sigma^2}x^2\right)$
Velocity Marginal	$p_{\text{stat}}(v) = \sqrt{\frac{a}{\pi\sigma^2}} \exp\left(-\frac{a}{\sigma^2}v^2\right)$
Fluctuation-Dissipation	$\sigma^2 = 2ak_BT$

Table 3: Entropy and Relaxation

Quantity	Formula
Gibbs Entropy	$S(t) = \frac{1}{2}\ln\left[(2\pi e)^2 \det\Sigma(t)\right]$
Entropy Production	$\frac{dS}{dt} = \frac{\sigma^2}{2}\frac{\Sigma_{22}-\Sigma_{22}^{\text{stat}}}{\Sigma_{22}^{\text{stat}}\Sigma_{22}} \geq 0$
Stationary Entropy	$S_{\text{stat}} = \frac{1}{2}\ln\left[(2\pi e)^2\frac{\sigma^4}{4a^2b}\right]$
Relaxation Time	$\tau_{\text{rel}} = \frac{1}{2\min\lvert\Re(r_i)\rvert}$

11 Special Cases and Asymptotics

11.1 No Damping Limit ($a \to 0$)

As $a \to 0$, the FPE becomes:

$$\frac{\partial p}{\partial t} = -v\frac{\partial p}{\partial x} + \frac{\partial}{\partial v}(bxp) + \frac{\sigma^2}{2}\frac{\partial^2 p}{\partial v^2}. \tag{74}$$

There is no stationary distribution (the energy grows without bound). The solution remains Gaussian but the variances grow as:

$$\Sigma_{11}(t) \sim \frac{\sigma^2 t^3}{3}, \quad \Sigma_{22}(t) \sim \sigma^2 t. \tag{75}$$

11.2 Overdamped Limit ($b \to 0$)

As $b \to 0$, the FPE becomes the Ornstein-Uhlenbeck process for velocity:

$$\frac{\partial p}{\partial t} = -v\frac{\partial p}{\partial x} + \frac{\partial}{\partial v}(avp) + \frac{\sigma^2}{2}\frac{\partial^2 p}{\partial v^2}. \tag{76}$$

The position variance grows diffusively:

$$\Sigma_{11}(t) \sim \frac{\sigma^2}{a^2}t. \tag{77}$$

11.3 Strong Damping Limit ($a \gg \sqrt{b}$)

In the strong damping limit, the FPE reduces to the Smoluchowski equation for position:

$$\frac{\partial p}{\partial t} = \frac{1}{a}\frac{\partial}{\partial x}(bxp) + \frac{\sigma^2}{2a^2}\frac{\partial^2 p}{\partial x^2}. \tag{78}$$

This describes overdamped Brownian motion in a harmonic potential.

12 Conclusion

We have provided a complete analytical treatment of the Fokker-Planck equation for the stochastic damped harmonic oscillator. The key results are:

1. **Exact Time-Dependent Solution:** The solution remains Gaussian for all times, with mean vector and covariance matrix given explicitly for all damping regimes.

2. **Complete Covariance Structure:** We have derived explicit formulas for $\Sigma_{11}(t)$, $\Sigma_{12}(t)$, and $\Sigma_{22}(t)$, revealing the finite-time correlation between position and velocity.

3. **Stationary Solution:** The stationary distribution is Gaussian with covariance:

   $$\Sigma_{\text{stat}} = \begin{pmatrix} \frac{\sigma^2}{2ab} & 0 \\ 0 & \frac{\sigma^2}{2a} \end{pmatrix},$$

   confirming the equipartition theorem.

4. **Entropy Dynamics:** We have analyzed the approach to equilibrium through entropy production:
$$\frac{dS}{dt} = \frac{\sigma^2}{2}\frac{\Sigma_{22} - \Sigma_{22}^{\text{stat}}}{\Sigma_{22}^{\text{stat}}\Sigma_{22}} \geq 0.$$
5. **Alternative Coordinate Systems:** We have explored energy-angle variables, normal coordinates, and complex representations, providing deeper physical insights.

These results are fundamental for understanding stochastic relaxation phenomena in physics, chemistry, and biology. The time-dependent Gaussian solution is essential for applications in stochastic control, filtering theory, and the analysis of experimental data from single-particle tracking and other noisy systems.

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A Derivation of the Fokker-Planck Equation

We provide a detailed derivation of the FPE (6) from the Itô SDE (2). For a function $\phi(x, v)$, Itô's lemma gives:

$$d\phi = \left(v\frac{\partial \phi}{\partial x} - (av + bx)\frac{\partial \phi}{\partial v} + \frac{\sigma^2}{2}\frac{\partial^2 \phi}{\partial v^2}\right) dt + \sigma\frac{\partial \phi}{\partial v}dW. \tag{A1}$$

Taking expectations and using the definition of the probability density:

$$\frac{d}{dt}\langle\phi\rangle = \left\langle v\frac{\partial \phi}{\partial x} - (av + bx)\frac{\partial \phi}{\partial v} + \frac{\sigma^2}{2}\frac{\partial^2 \phi}{\partial v^2}\right\rangle. \tag{A2}$$

Integration by parts yields:

$$\int\int \phi\frac{\partial p}{\partial t}dxdv = \int\int \phi\left[-v\frac{\partial p}{\partial x} + \frac{\partial}{\partial v}\left((av + bx)p\right) + \frac{\sigma^2}{2}\frac{\partial^2 p}{\partial v^2}\right] dxdv. \tag{A3}$$

Since ϕ is arbitrary, this proves (6). □

B Solving the Lyapunov Equation

The Lyapunov equation (16) can be solved systematically. The solution for the covariance matrix elements satisfies:

$$\dot{\Sigma}_{11} = 2\Sigma_{12}, \tag{B1}$$

$$\dot{\Sigma}_{12} = \Sigma_{22} - a\Sigma_{12} - b\Sigma_{11}, \tag{B2}$$

$$\dot{\Sigma}_{22} = -2a\Sigma_{22} - 2b\Sigma_{12} + \sigma^2. \tag{B3}$$

This is a system of linear ODEs. Taking Laplace transforms or solving via the matrix exponential gives the results presented in Section 4. □

C Derivation of Entropy Production

From (48):

$$\frac{dS}{dt} = \frac{1}{2}\frac{d}{dt}\ln\det\Sigma = \frac{1}{2}\mathrm{Tr}\left(\Sigma^{-1}\dot{\Sigma}\right). \tag{79}$$

Using (16):

$$\frac{dS}{dt} = \frac{1}{2}\mathrm{Tr}\left(\Sigma^{-1}(A\Sigma + \Sigma A^T + BB^T)\right). \tag{80}$$

$$= \frac{1}{2}\mathrm{Tr}(A + A^T + \Sigma^{-1}BB^T). \tag{81}$$

Since $\mathrm{Tr}(A) = -a$:

$$\frac{dS}{dt} = -a + \frac{1}{2}\mathrm{Tr}(\Sigma^{-1}BB^T). \tag{82}$$

Now $\mathrm{Tr}(\Sigma^{-1}BB^T) = \sigma^2[\Sigma^{-1}]_{22} = \sigma^2\frac{\Sigma_{11}}{\det\Sigma}$.
Thus:

$$\frac{dS}{dt} = -a + \frac{\sigma^2}{2}\frac{\Sigma_{11}}{\Sigma_{11}\Sigma_{22} - \Sigma_{12}^2}. \tag{83}$$

In the stationary state, $\Sigma_{22}^{\mathrm{stat}} = \sigma^2/(2a)$, so $\sigma^2/2 = a\Sigma_{22}^{\mathrm{stat}}$. This gives the result in Section 7.2. □

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